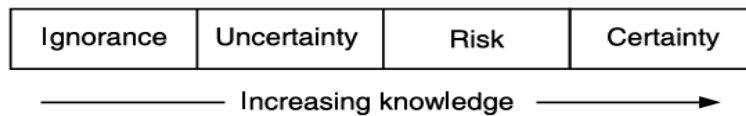


DECISION THEORY

The success or failure that an individual or organization experiences, depends to a large extent, on the ability of making acceptable decisions on time. To arrive at such a decision, a decision-maker needs to enumerate feasible and viable courses of action (alternatives or strategies), the projection of consequences associated with each course of action, and a measure of effectiveness (or an objective) to identify the best course of action.

Decision theory is both descriptive and prescriptive business modelling approach to classify the degree of knowledge and compare expected outcomes due to several courses of action. The degree of knowledge is divided into four categories: complete knowledge (i.e. certainty), ignorance, risk and uncertainty as shown



The Three Environments of Decision-Making

From a practical standpoint, the "way" you approach a decision depends entirely on the amount of information available. Decisions are made in one of three environments:

Decision Environment	Definition	Core Approach	Example
Under Certainty	You have complete data and know exactly what the outcome of each choice will be.	Choose the alternative that yields the highest absolute return or benefit.	Purchasing a government bond with a fixed, guaranteed interest rate.
Under Risk	You do not know the exact outcome, but you know the mathematical probabilities of each scenario.	Calculate the Expected Monetary Value (EMV) or utility by multiplying outcomes by their likelihood.	Launching a product when market research shows a 70% chance of high success and a 30% chance of failure.
Under Uncertainty	You have no idea what the future holds and cannot accurately assign probabilities to potential outcomes.	Apply logical criteria based on risk tolerance (e.g., Maximax for optimists or Minimax for pessimists).	Investing in a highly disruptive, entirely new technology sector with zero historical data.

Example: A company deciding how many units to produce

A company must decide whether to produce 100, 200, or 300 units of a new product.

The profit depends on market demand.

Decision	Low demand	Medium demand	High demand
Produce 100 units	₹40,000	₹40,000	₹40,000
Produce 200 units	₹20,000	₹70,000	₹80,000
Produce 300 units	−₹20,000	₹50,000	₹1,20,000

Now the three decision situations can be explained using this same example.

1. Decision-making under Certainty

Meaning

Under **certainty**, the decision-maker **knows exactly which state of nature will occur**.

Example

Suppose the company has reliable confirmed orders showing that the demand will be **high**.

Therefore, the company knows:

High demand will occur with certainty.

Looking at the table:

- Produce 100 → ₹40,000
- Produce 200 → ₹80,000
- Produce 300 → ₹1,20,000

The company can choose **300 units**, because the future state is known.

Key point

Certainty = State of nature is known

Question:

"What will happen?"

Answer: We know what will happen.

2. Decision-making under Risk

Meaning: Under **risk**, the possible future states are known **and their probabilities are known**.

Suppose market research gives:

Demand	Probability
Low	0.20
Medium	0.50
High	0.30

Now calculate the **Expected Monetary Value (EMV)**.

Produce 100

$$\text{EMV (100)} = 0.20(40,000) + 0.50(40,000) + 0.30(40,000) = ₹40,000$$

Produce 200

$$\text{EMV (200)} = 0.20(20,000) + 0.50(70,000) + 0.30(80,000) = ₹59,000$$

Produce 300

$$\text{EMV (300)} = 0.20(-20,000) + 0.50(50,000) + 0.30(1,20,000) = ₹46,000$$

So the EMVs are:

Decision	EMV
100 units	₹40,000
200 units	₹59,000
300 units	₹46,000

Under the **EMV criterion**, the decision corresponding to ₹59,000 has the highest expected monetary value.

Key point

Risk = Possible outcomes known + probabilities known

Question:

"What might happen, and how likely is each outcome?"

3. Decision-making under Uncertainty

Meaning

Under **uncertainty**, the possible states of nature may be known, but their **probabilities are not known or cannot be reliably estimated**.

Suppose the company knows that demand could be:

- Low
- Medium
- High

But it has **no reliable information about the probabilities**.

So the company cannot say:

"Low demand has probability 0.20." or "High demand has probability 0.30."

Therefore, EMV cannot be calculated reliably.

The company must use decision criteria such as:

- **Maximax criterion** – optimistic approach

- **Maximin criterion** – conservative approach
- **Minimax Regret criterion**
- **Laplace criterion** – assume equal likelihood when appropriate

Key point

Uncertainty = Possible outcomes known, probabilities unknown

Question:

"What might happen, but how likely is each outcome?"

Answer: We don't know the probabilities.

1. INTRODUCTION TO DECISION MAKING

1.1 Meaning of Decision Making

Decision making is the process of selecting the most appropriate course of action from a set of available alternatives to achieve a desired objective.

In real-life situations, a decision-maker may not know exactly what will happen in the future. Different future conditions may lead to different outcomes. Decision theory provides mathematical and logical tools for choosing among alternatives.

Simple example

A company wants to decide whether to develop a new software product.

It has three alternatives:

- Develop a small system
- Develop a medium system
- Develop a large system

The profit will depend on future market demand:

- Low demand
- Medium demand
- High demand

The company must decide which alternative to choose.

2. Importance of Decision Making

Decision making is important because organizations and individuals frequently have to make choices when resources are limited and future outcomes are uncertain.

Major applications

1. Resource allocation

- How much money, time, computing power, or manpower should be allocated?

2. Project selection

- Which software or engineering project should be undertaken?

3. Production planning

- How many units should be produced?

4. Investment decisions

- Which investment option should be selected?

5. Technology selection

- Which hardware, software, cloud platform, or technology should be adopted?

6. Capacity planning

- How much server capacity should be installed?

7. Risk management

- What action should be taken when future outcomes are uncertain?

8. Artificial intelligence and machine learning

- Selecting an action based on predicted outcomes and associated costs.

3. Elements of a Decision Problem

A decision problem generally consists of four important elements:

Actions + States of Nature + Payoffs + Decision Criterion

The three fundamental elements are **actions, states of nature, and payoffs**.

3.1 Actions / Alternatives

An **action** is a possible decision or course of action available to the decision-maker.

They are usually denoted by $A_1, A_2, \dots, A_m$

Example

A software company has three possible actions:

A_1 =Develop basic software

A_2 =Develop standard software

A_3 =Develop advanced software

The company must select one of these actions.

3.2 States of Nature

A **state of nature** represents a possible future condition that affects the outcome of a decision but is **not controlled by the decision-maker**.

They are usually denoted by $S_1, S_2, \dots, S_n$

For example:

S1=Low demand

S2=Medium demand

S3=High demand

The company cannot control whether demand will be low, medium, or high.

Important distinction

Action → controlled by decision-maker

State of nature → not controlled by decision-maker

3.3 Payoff

A **payoff** is the result obtained when a particular action is chosen and a particular state of nature occurs.

Payoffs may represent:

- Profit
- Revenue
- Cost savings
- Performance
- Production
- Time saved

For a profit problem, a higher payoff is generally preferred.

For a cost problem, a lower payoff is generally preferred.

4. Decision Table

A **decision table** presents the actions, states of nature, and corresponding payoffs in a systematic form.

Example

A company has three production alternatives.

Action	Low demand	Medium demand	High demand
A1: Small plant	₹40,000	₹40,000	₹40,000
A2: Medium plant	₹20,000	₹70,000	₹80,000
A3: Large plant	−₹20,000	₹50,000	₹1,20,000

Here:

- **Actions:** Small, Medium, Large plant
- **States of nature:** Low, Medium, High demand
- **Entries:** Payoffs

This table is called a **payoff table** or **payoff matrix**.

5. Payoff Matrix

A **payoff matrix** is a mathematical representation of the outcomes associated with different actions and states of nature.

In general: $P = [P_{ij}]$

Where

P_{ij} = payoff obtained by choosing A_i when state S_j occurs.

For example:
$$P = \begin{bmatrix} 40000 & 40000 & 40000 \\ 20000 & 70000 & 80000 \\ -20000 & 50000 & 120000 \end{bmatrix}$$

Rows represent **actions** and columns represent **states of nature**.

6. Types of Decision Making

Decision problems are commonly classified into:

1. Decision making under certainty

The state of nature is known with certainty.

2. Decision making under risk

The possible states are known and their probabilities are known.

3. Decision making under uncertainty

The possible states are known, but their probabilities are unknown.

This chapter focuses particularly on **uncertainty and risk**.

PART A: DECISION MAKING UNDER UNCERTAINTY

Meaning of Decision Making Under Uncertainty

A decision is made under **uncertainty** when:

1. There are several possible states of nature.
2. The decision-maker does not know which state will occur.
3. The probabilities of the states are **not known**.

For example, a company is launching a completely new product and has no reliable information about future demand.

Possible demand:

- Low
- Medium
- High

But probabilities cannot be estimated reliably.

Therefore, probability-based methods such as EMV cannot directly be applied.

Instead, several **decision criteria** are used.

Decision Criteria Under Uncertainty

The major criteria are:

1. **Optimism (Maximax or Minimin) criterion**
2. **Pessimism (Maximin or Minimax) criterion**
3. **Minimax Regret criterion**
4. **Laplace criterion** (Equal probabilities)
5. **Hurwicz criterion** (Coefficient of optimism)

We will use the following payoff matrix throughout example:

Alternative	Low demand	Medium demand	High demand
A1 Small	40	40	40
A2 Medium	20	70	80
A3 Large	−20	50	120

Payoffs are in ₹ **thousands**.

Maximax Criterion

Maximax means: Maximum of the maximum payoffs

It represents an **optimistic approach**.

The decision-maker assumes that the most favourable state of nature will occur.

Step 1: Find the maximum payoff for each action

Action	Payoffs	Row maximum
A1	40, 40, 40	40
A2	20, 70, 80	80
A3	−20, 50, 120	120

Step 2

Select the largest row maximum:

Max (40, 80, 120) = 120

Therefore, according to **Maximax**, select: A3

Interpretation

The decision-maker focuses on the **best possible payoff**.

Maximax = Best of the best

Maximin Criterion

Maximin means: Maximum of the minimum payoffs

It represents a **conservative approach**.

The decision-maker considers the worst outcome for each action and then selects the action whose worst outcome is the best among the alternatives.

Step 1: Find minimum payoff in each row

Action	Payoffs	Row minimum
A1	40, 40, 40	40
A2	20, 70, 80	20
A3	-20, 50, 120	-20

Step 2

Select the largest minimum: $\max(40, 20, -20) = 40$

Therefore: A1

Interpretation

The decision-maker asks: "If the worst possible outcome occurs, which decision protects me best?"

Maximin = Best of the worst

Maximax vs Maximin

Criterion	Approach	Calculation
Maximax	Optimistic	Maximum of row maxima
Maximin	Conservative	Maximum of row minima

For the example:

Maximax=A3 Maximin=A1

The different answers occur because the two criteria have different attitudes toward uncertainty.

Minimax Regret Criterion

Regret is the loss caused by not selecting the best action after the actual state of nature becomes known.

It is also called:

- Opportunity loss

- Regret

For each state of nature:

Regret=Best payoff in that state–Actual payoff

Constructing the Regret Matrix

Original payoff matrix:

Action	Low	Medium	High
A1	40	40	40
A2	20	70	80
A3	–20	50	120

Step 1: Find the maximum payoff in each column

- Low demand → 40
- Medium demand → 70
- High demand → 120

Step 2: Calculate regret

For Low: $40-40=0$ $40-20=20$ $40-(-20) = 60$

For Medium: $70-40=30$ $70-70=0$ $70-50=20$

For High: $120-40=80$ $120-80=40$ $120-120=0$

Therefore:

Action	Low	Medium	High	Maximum regret
A1	0	30	80	80
A2	20	0	40	40
A3	60	20	0	60

Step 3: Apply Minimax Regret

Choose the **minimum of the maximum regrets**: $\min (80, 40, 60) = 40$

Therefore: A2

Minimax Regret = Minimize the maximum possible regret.

Laplace Criterion

Meaning: The **Laplace criterion** is used when probabilities are unknown and the decision-maker assumes that all states of nature are **equally likely**.

If there are n states $P(S_i) = \frac{1}{n}$

The average payoff is $L(A_i) = \frac{\sum_{j=1}^n P_{ij}}{n}$

Choose the action with the **highest average payoff** for a payoff-maximization problem.

Example

Using our payoff matrix:

$$L(A_1) = \frac{40+40+40}{3} = 40$$

$$L(A_2) = \frac{20+70+80}{3} = 56.67$$

$$L(A_3) = \frac{-20+50+120}{3} = 50$$

Therefore: A2, Because $56.67 > 50 > 40$

Interpretation: Laplace says: "Since I have no probability information, treat the states equally."

Important note: Equal probabilities are an **assumption of the Laplace criterion**, not evidence that the states are actually equally likely.

Hurwicz Criterion

This criterion suggests that a decision-maker should be neither completely optimistic nor pessimistic and, therefore, must display a mixture of both. Hurwicz, who suggested this criterion, introduced the idea of a coefficient of optimism (denoted by α) to measure the decision-maker's degree of optimism. This coefficient lies between 0 and 1, where 0 represents a complete pessimistic attitude about the future and 1 a complete optimistic attitude about the future. Thus, if α is the coefficient of optimism, then $(1 - \alpha)$ will represent the coefficient of pessimism.

The Hurwicz approach suggests that the decision-maker must select an alternative that maximizes

$$H (\text{Criterion of realism}) = \alpha (\text{Maximum in column}) + (1 - \alpha) (\text{Minimum in column})$$

The working method is summarized as follows:

- Decide the coefficient of optimism α (alpha) and then coefficient of pessimism $(1 - \alpha)$.
- For each decision alternative select the largest and lowest payoff value and multiply these with α and $(1 - \alpha)$ values, respectively. Then calculate the weighted average, H by using above formula.
- Select an alternative with best weighted average payoff value.

Meaning: The **Hurwicz criterion** combines optimistic and pessimistic viewpoints.

It uses a coefficient of optimism:

$$\alpha, \quad 0 \leq \alpha \leq 1$$

The decision value is:

$$H(A_i) = \alpha(\text{Maximum payoff}) + (1 - \alpha)(\text{Minimum payoff})$$

For a profit/payoff problem, select the action with the **largest Hurwicz value**.

Example

Suppose:

$$\alpha = 0.6$$

Therefore:

$$1 - \alpha = 0.4$$

For A_1

Maximum = 40

Minimum = 40

$$\begin{aligned} H(A_1) &= 0.6(40) + 0.4(40) \\ &= 40 \end{aligned}$$

For A_2

Maximum = 80

Minimum = 20

$$\begin{aligned} H(A_2) &= 0.6(80) + 0.4(20) \\ &= 48 + 8 = 56 \end{aligned}$$

For A_3

Maximum = 120

Minimum = -20

$$\begin{aligned} H(A_3) &= 0.6(120) + 0.4(-20) \\ &= 72 - 8 = 64 \end{aligned}$$

Therefore:

$$\boxed{A_3}$$

15. Interpretation of the Hurwicz Coefficient

$$0 \leq \alpha \leq 1$$

If $\alpha = 1$

The decision-maker is completely optimistic.

$$H(A_i) = \max(P_{ij})$$

This becomes **Maximax**.

If $\alpha = 0$

The decision-maker is completely pessimistic.

$$H(A_i) = \min(P_{ij})$$

For selecting the largest value, this becomes **Maximin**.

Thus:

Hurwicz lies between optimism and pessimism

Summary of Criteria Under Uncertainty

Criterion	Basic idea	Rule for payoff
Maximax	Most optimistic	Maximum of row maxima
Maximin	Most conservative	Maximum of row minima
Minimax Regret	Minimize worst regret	Minimum of row maximum regrets
Laplace	Equal likelihood	Maximum average payoff
Hurwicz	Weighted optimism/pessimism	Maximum weighted payoff

Practice Problems:

1. The following matrix gives the payoff (in Rs) of different strategies (alternatives) S_1 , S_2 and S_3 against conditions (events) N_1 , N_2 , N_3 and N_4 .

Strategy	State of Nature			
	N_1	N_2	N_3	N_4
S_1	4,000	– 100	6,000	18,000
S_2	20,000	5,000	400	0
S_3	20,000	15,000	– 2,000	1,000

Indicate the decision taken under the following approaches: (i) Pessimistic, (ii) Optimistic, (iii) Equal probability, (iv) Regret, (v) Hurwicz criterion, the degree of optimism being 0.7.

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Indicate the decision taken under the following approaches: (i) Pessimistic, (ii) Optimistic, (iii) Equal probability, (iv) Regret, (v) Hurwicz criterion, the degree of optimism being 0.7.

2. In a toy manufacturing company, suppose the product acceptance probabilities are not known but the following data is known:

Product Acceptance	Anticipated First Year Profit ('000 Rs) Product Line		
	Full	Partial	Minimal
Good	8	70	50
Fair	50	45	40
Poor	-25	-10	0

Determine the optimal decision under each of the following decision criteria and show how you arrived at it: (a) Maximax, (b) Maximin, (c) Equal likelihood and (d) Minimax regret?

3. The following is a payoff (in rupees) table for three strategies and two states of nature:

Strategy	State of Nature	
	N_1	N_2
S_1	40	60
S_2	10	-20
S_3	-40	150

Select a strategy using each of the following decision criteria: (a) Maximax, (b) Minimax regret, (c) Maximin, (d) Minimum risk, assuming equiprobable states.

4. Mr Sethi has Rs 10,000 to invest in one of three options: A, B or C. The return on his investment depends on whether the economy experiences inflation, recession, or no change at all. The possible returns under each economic condition are given below:

Strategy	State of Nature		
	Inflation	Recession	No Change
A	2,000	1,200	1,500
B	3,000	800	1,000
C	2,500	1,000	1,800

What should he decide, using the pessimistic criterion, optimistic criterion, equally likely criterion and regret criterion?

- (i) S_2 , (ii) S_2 or S_3 , (iii) S_3 , (iv) S_1 , (v) S_2
- (a) Full, (b) Minimal, (c) Full or partial, (d) Partial
- (a) S_3 ; Rs 150 (b) S_3 ; Rs 80 (c) S_1 ; Rs 40 (d) S_3 ; Rs 55
- Choose A: Rs 120, Choose B: Rs 300, Choose C: Rs 176.6, Choose C: Rs 50